

SynkroBoard

1. Problem Statement

Modern organizations, distributed teams, project managers, and executives struggle with fragmented workflows, lack of transparent task accountability, scattered communication channels, and rigid project management tools. Existing platforms often disconnect high-level strategic tracking (RACI matrix, milestones, team availability) from daily execution (Kanban boards, subtasks, real-time discussions), resulting in missed deadlines, resource bottlenecks, unmonitored team absences, and inefficient team collaboration.

2. Our Solution

SynkroBoard provides an all-in-one enterprise task and workspace management ecosystem that unifies strategic project governance with real-time operational execution. It includes a responsive Web Application and cross-platform Mobile Applications (iOS & Android) for on-the-go productivity and team chat, powered by a centralized NestJS backend with real-time WebSocket synchronization, role-based access control, intelligent absence-aware task scheduling, and comprehensive auditability.

3. Project Overview

SynkroBoard is an **All-in-one** collaborative project and workspace management platform tailored for modern enterprises, agile teams, and fast-moving organizations. It bridges the gap between project planning, task execution, team communication, and productivity analytics. The system features multi-view project visualization (Interactive Kanban, RACI Matrix, Gantt Timelines, Eisenhower Matrix, Day Planner/Calendar), contextual real-time team chat, automated leave/absence tracking, and centralized role-based Dashboards (Super Admin, Admin, and Employee) for complete organizational oversight.

4. Goals

Project Managers & Team Leaders

- Provide complete visibility over project timelines, task lifecycles, and team velocity in real time.
- Eliminate ambiguity in ownership with built-in RACI responsibility matrices and automated milestone tracking.
- Optimize team workload and automatically adjust task deadlines based on member leaves and availability.
- Accelerate project delivery through agile Kanban boards with customizable workflows and WIP limits.

Team Members & Employees

- Empower employees with an intuitive "Today's Focus" / "My Day" dashboard for effortless daily task prioritization.
- Streamline task execution with rich subtask checklists, markdown descriptions, file attachments, and activity timelines.

- Enhance workplace communication with integrated, instant team chat, group channels, and real-time socket notifications.
- Simplify absence management with 1-click leave requests and transparent holiday schedules.

Executives & Workspace Administrators (Super Admins)

- Maintain enterprise-grade data security with multi-tenant workspace isolation and granular role-based permissions (RBAC).
- Gain actionable organizational insights through executive-level productivity analytics and cross-project health metrics.
- Seamlessly govern user onboarding, workspace memberships, and system-wide configurations from a single centralized hub.
- Foster a culture of accountability, transparency, and operational excellence across the entire enterprise.

5. Vision

To redefine modern workplace collaboration by building a unified, intelligent, and human-centric management platform where every team member is empowered to focus, execute with clarity, and achieve peak productivity without friction.

6. Our Values

- **Clarity & Accountability:** Ensure every project role, task ownership, and milestone is crystal clear through structured RACI modeling.
- **Real-time Agility:** Deliver instantaneous state synchronization across Web and Mobile via live WebSockets.
- **Employee Well-being & Balance:** Respect human capacity by integrating absence awareness directly into project schedules and deadlines.
- **Data Privacy & Security:** Guarantee enterprise-level data segregation through strict multi-tenancy and role-based access control.
- **User-Centric Simplicity:** Craft clean, modern, and friction-free user experiences that teams genuinely love using every day.
- **Continuous Innovation:** Empower teams with versatile views (Kanban, Gantt, Eisenhower, Calendar) tailored to diverse working styles.
- **Collaborative Synergy:** Unify execution with contextual communication so discussions happen where work is being done.

7. What the System Does

Area	Purpose
Workspace Management	Create, configure, and isolate multi-tenant workspaces with custom settings and member role permissions.
Project Management	Organize projects with categories, status workflows, budget tracking, member assignments, and health metrics.
Kanban Board System	Visual workflow management with customizable stages/columns, WIP limits, color tagging, card archiving, and reordering.
Task Lifecycle & Checklists	Comprehensive task tracking with priorities, due dates, estimates, subtask checklists, and file attachments.
RACI Matrix Governance	Formal assignment and auditing of Responsible, Accountable, Consulted, and Informed stakeholders per task.
Gantt & Timeline Scheduling	Interactive project schedules, dependency tracking, duration planning, and milestone monitoring.
Eisenhower Matrix Planning	4-Quadrant prioritization (Urgent/Important) for rapid decision making and high-impact task execution.
Calendar & Day Planner	Centralized team calendar combining project deadlines, personal "Today's Focus" schedules, and holiday events.
Leave & Absence Management	Submit and approve employee leaves (Sick, Casual, Annual) with automated task timeline recalculation.
Real-Time Team Chat	Contextual direct messaging and group channels with live typing indicators, read receipts, and emoji reactions.
Alerts & Push Notifications	Real-time WebSocket alerts and mobile push notifications for task assignments, mentions, and approaching deadlines.
Role-Based Dashboards	Dedicated analytical views for Super Admins (system health), Admins (team progress), and Employees (personal focus).
User & Role Administration	Comprehensive user lifecycle management, invitation flows, status toggles, and granular role assignments.

8. Platform Ecosystem

- **Web Application** – A high-performance, responsive desktop application for workspace administrators, project managers, and collaborative power users.
- **Mobile Application** – A native iOS & Android mobile application for on-the-go task updates, real-time chat, leaves, calendar, and push notifications.
- **Centralized Backend API** – A robust RESTful and WebSocket server providing real-time synchronization, enterprise security (JWT/RBAC), background scheduling, and automated workflows.

9. Key Features

- Multi-Tenant Workspace & Project Management
- Interactive Kanban Boards with Custom Columns & Archiving
- Task Management with Subtask Checklists & File Attachments
- RACI Matrix Responsibility Governance
- Interactive Gantt Chart & Timeline View
- Eisenhower Priority Matrix Quadrants
- Unified Calendar & Personal Day Planner ("Today's Focus")
- Employee Leave & Absence Tracking with Auto-Schedule Sync
- Real-Time WebSocket Chat (Direct & Group Channels)
- Instant Live Notifications & Activity Auditing
- Role-Based Dashboards (Super Admin, Admin, Employee)
- Granular Role-Based Access Control (RBAC) & Member Invitations
- Cross-Platform Web & Mobile (iOS & Android) Parity